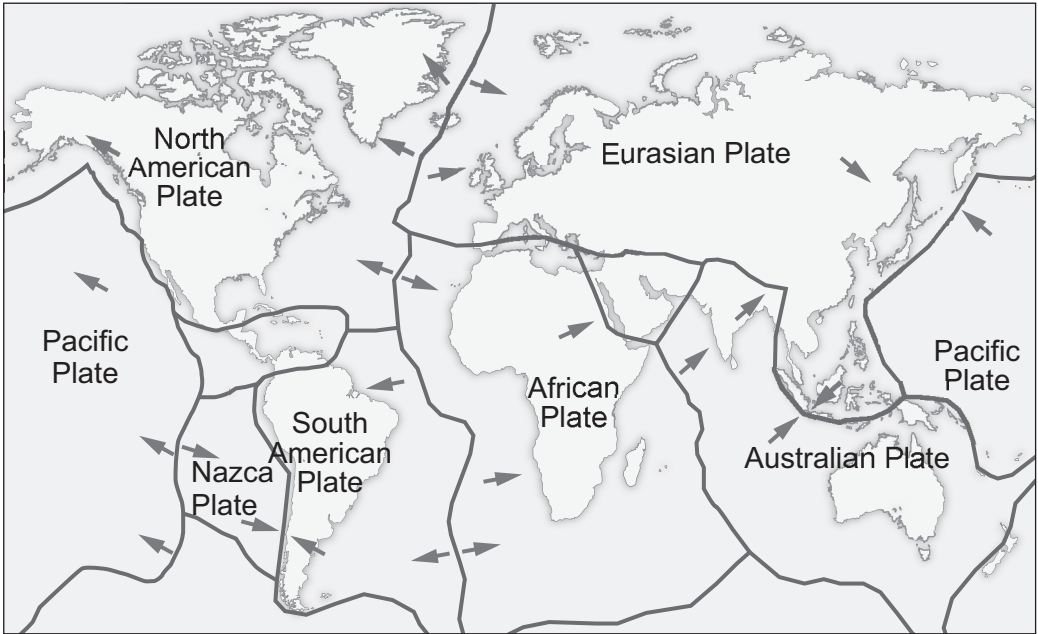


11. The following diagram shows some of the Earth's tectonic plates and the direction in which they move.



(a) The boundary between the Nazca and South American plates is a destructive plate boundary. Describe what happens at a destructive boundary. [2]

.....

.....

.....

(b) Draw a cross (✕) on the diagram to show a constructive plate boundary. Describe what happens at this boundary. [2]

.....

.....

.....

(c) State **one** effect of plates sliding past each other. [1]

.....

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1

H

1

Hydrogen

1H Hydrogen 1																											
7 Li Lithium 3	9 Be Beryllium 4																										
23 Na Sodium 11	24 Mg Magnesium 12																										
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30																
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	127 I Iodine 53	128 Te Tellurium 52	131 Xe Xenon 54										
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86										
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																									

Key

4He Helium 2																								
20Ne Neon 10	19F Fluorine 9	16O Oxygen 8	14N Nitrogen 7	12C Carbon 6	11B Boron 5																			
40Ar Argon 18	35.5Cl Chlorine 17	32S Sulfur 16	31P Phosphorus 15	28Si Silicon 14	27Al Aluminium 13																			
84Kr Krypton 36	80Br Bromine 35	79Se Selenium 34	75As Arsenic 33	73Ge Germanium 32	70Ga Gallium 31	65Zn Zinc 30																		
131Xe Xenon 54	127I Iodine 53	128Te Tellurium 52	122Sb Antimony 51	119Sn Tin 50	115In Indium 49	112Cd Cadmium 48																		
222Rn Radon 86	210At Astatine 85	210Po Polonium 84	209Bi Bismuth 83	207Pb Lead 82	204Tl Thallium 81	201Hg Mercury 80																		

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3410U10-1



S19-3410U10-1

WEDNESDAY, 12 JUNE 2019 – MORNING

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and
Essential Resources

FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper
 you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen
 or correction fluid.

Write your name, centre number and candidate number
 in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.
 If you run out of space, use the additional page at the back
 of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 7(a) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	7	
3.	8	
4.	6	
5.	6	
6.	6	
7.	10	
8.	10	
9.	7	
10.	7	
11.	6	
Total	80	

3410U101
01

JUN193410U10101

Examiner
only

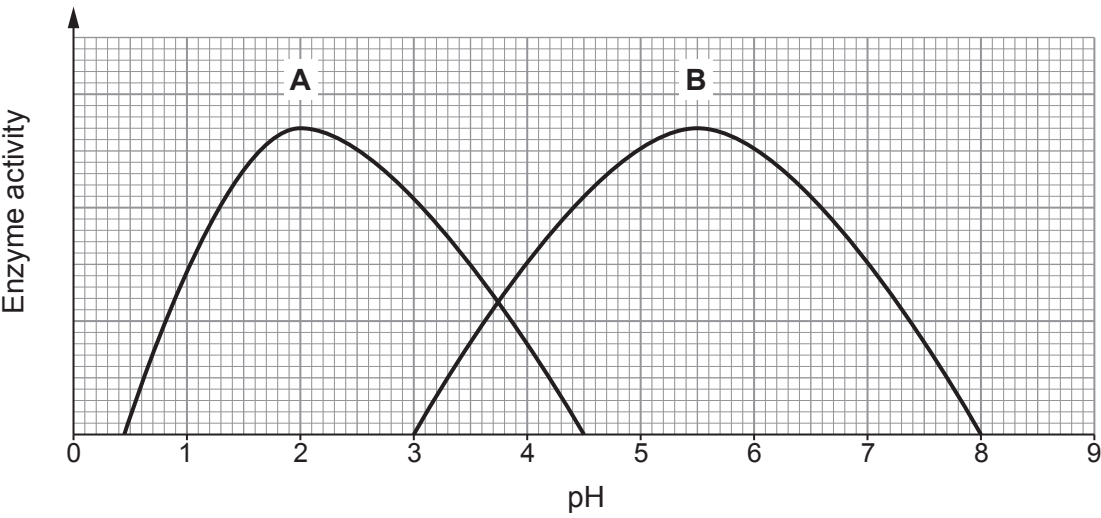
11. (a) Enzymes are biological catalysts. State what is meant by the term *catalyst*. [2]

.....

.....

.....

(b) The following graphs show how the activity of two enzymes, **A** and **B**, varies with pH.



(i) Use the graphs to compare the activities of the two enzymes. [2]

.....

.....

.....

(ii) Enzyme **C** is found in saliva. It works between pH 5 and pH 9 but is best at a neutral pH. **Sketch on the grid above** how the activity varies with pH. [2]

6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1

H

Hydrogen

1

1 H Hydrogen 1																								

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410U10-1



Z22-3410U10-1

FRIDAY, 17 JUNE 2022 – AFTERNOON

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and
Essential Resources

FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **8** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	3	
3.	6	
4.	8	
5.	12	
6.	8	
7.	10	
8.	6	
9.	5	
10.	9	
11.	6	
Total	80	

3410U101
01

JUN223410U10101

Examiner
only

11. (a) The Earth’s early atmosphere contained large amounts of water vapour and carbon dioxide. Explain why the amounts of water vapour and carbon dioxide decreased over geological time. [4]

Water vapour

.....

.....

.....

Carbon dioxide

.....

.....

.....

- (b) State the percentages of nitrogen and oxygen in the present atmosphere. [2]

Nitrogen %

Oxygen %

6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2

Group

3

4

5

6

7

0

1 H Hydrogen 1										4 He Helium 2																																																									
7 Li Lithium 3		9 Be Beryllium 4																		19 F Fluorine 9		16 O Oxygen 8		14 N Nitrogen 7		12 C Carbon 6		11 B Boron 5																		20 Ne Neon 10																					
23 Na Sodium 11		24 Mg Magnesium 12																		35.5 Cl Chlorine 17		32 S Sulfur 16		31 P Phosphorus 15		28 Si Silicon 14		27 Al Aluminium 13																		40 Ar Argon 18																					
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30																		80 Br Bromine 35		79 Se Selenium 34		75 As Arsenic 33		73 Ge Germanium 32		70 Ga Gallium 31																		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48																		127 I Iodine 53		128 Te Tellurium 52		122 Sb Antimony 51		119 Sn Tin 50		115 In Indium 49																		131 Xe Xenon 54	
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80																		210 At Astatine 85		210 Po Polonium 84		209 Bi Bismuth 83		207 Pb Lead 82		204 Tl Thallium 81																		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																																																															

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3410U10-1



FRIDAY, 16 JUNE 2023 – MORNING

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and
Essential Resources

FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in Question 5(b).
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	8	
3.	7	
4.	10	
5.	9	
6.	6	
7.	5	
8.	8	
9.	9	
10.	11	
Total	80	



JUN233410U10101